Question:
Consider a 128-bit AES plaintext block and a 128-bit AES key.

1. Explain the main steps performed during an AES encryption round.
2. Why is **MixColumns omitted in the final AES round**?
3. If AES-128 uses **10 rounds**, identify which transformations are performed in the initial round, middle rounds, and final round.

Solution:

AES-128 encryption consists of an **initial transformation followed by 10 rounds**.

1. Initial Round:

- **AddRoundKey** is performed.
- The plaintext is XORed with the first round key.

2. Middle Rounds (Rounds 1–9):
Each round consists of:

1. **SubBytes** — Each byte is replaced using the AES S-box.
2. **ShiftRows** — Rows of the state matrix are cyclically shifted.
3. **MixColumns** — Bytes in each column are mathematically mixed.
4. **AddRoundKey** — The state is XORed with the corresponding round key.

3. Final Round (Round 10):

1. **SubBytes**
2. **ShiftRows**
3. **AddRoundKey**

MixColumns is omitted from the final round because AES was specifically designed this way. Including it would not provide the intended additional security and would make the encryption/decryption structure unnecessarily more complex.

Therefore:

Initial round: AddRoundKey
Rounds 1–9: SubBytes → ShiftRows → MixColumns → AddRoundKey
Round 10: SubBytes → ShiftRows → AddRoundKey